

Fully Automated Wafer-Level Grating and Edge Coupling Measurement System for Silicon Photonics Integrated Circuits

Quan Yuan, Joe Frankel, and Divya Pratap

FormFactor Inc., 9100 SW Gemini Dr, Beaverton, OR 97008, USA

quan.yuan@formfactor.com

Abstract: We demonstrate a fully automated wafer-level optical probing platform achieving 8.9% GRR for grating-coupler wavelength sweeps and ± 0.03 – 0.06 dB edge-coupler repeatability. A 3D trajectory method enables direct wafer-level incident-angle measurement under automated operation.

1. Introduction

Silicon photonic integrated circuits (PICs) continue to scale in channel density, wafer size, and integration complexity for data center interconnects and co-packaged optics (CPO). As device complexity increases, wafer-level optical testing has become essential for process monitoring and yield validation [1]–[3]. However, achieving foundry-grade repeatability remains challenging due to mechanical instability, polarization sensitivity, and angular misalignment, which directly impact coupling efficiency and spectral accuracy.

Grating couplers (GCs) and edge couplers (ECs) represent complementary wafer-level optical interfaces. GC probing provides vertical access but exhibits strong sensitivity to incident angle and polarization state [4], [5]. EC probing offers lower insertion loss and broadband operation but requires sub-micron alignment precision and mechanical stability [6], [7]. For optical wafer acceptance testing (OWAT), measurement uncertainty must remain well below the process tolerance, typically requiring gauge repeatability and reproducibility (GRR) below 10%.

Our previous work demonstrated a fully automated wafer-level edge-coupling (EC) platform achieving <0.2 dB repeatability [8]. In this work, we extend the architecture to a unified GC and EC probing system incorporating mechanical loop optimization, stabilized polarization control, and curve-fitted peak extraction. The platform achieves <0.1 dB repeatability for both GC and EC measurements. In addition, the grating-coupler wavelength-sweep GRR is reduced to $<10\%$ based on a ± 0.08 dB tolerance target, satisfying typical wafer-level acceptance testing requirements.

2. System Overview

The platform is a fully automated wafer-level optical probing system supporting both GC and EC characterization on 200 mm and 300 mm silicon photonic wafers. The architecture integrates high-precision positioning, polarization control, tunable laser instrumentation, and automated data acquisition within a probe-station environment.

Experiments were performed on a FormFactor CM300xi-SiPh probe station. Optical alignment is achieved using a 6-DOF hexapod manipulator (PI H-811) combined with high-speed piezo nanopositioners (PI P-611.3). A tunable laser source (Keysight N7776C) provides O-band excitation. Polarization is controlled using a Keysight N7786C polarization controller. Spectral acquisition is performed using Keysight N7745C instrumentation with simultaneous alignment monitoring through a fused splitter and Newport 2103 power meter. Motion control and data logging are fully automated to support wafer-scale repeatability studies.

For GC testing, light is coupled through a 10.8° polished FAU probing 250 μm -pitch loopback structures. For EC testing, a periscope-structured FormFactor Pharos micro-lens assembly couples light into V-groove edge couplers. The unified configuration enables automated wavelength sweeps, 3D coupling trajectory reconstruction, and round-robin repeatability evaluation.

3. Grating Coupler Performance

3.1 Wavelength Sweep Repeatability

A single loop-back GC structure (O-band, MFD $\approx 9.2 \mu\text{m}$) was characterized using ten consecutive wavelength sweeps from 1280 nm to 1340 nm following alignment at 1310 nm. Peak power and peak wavelength were extracted from each sweep.

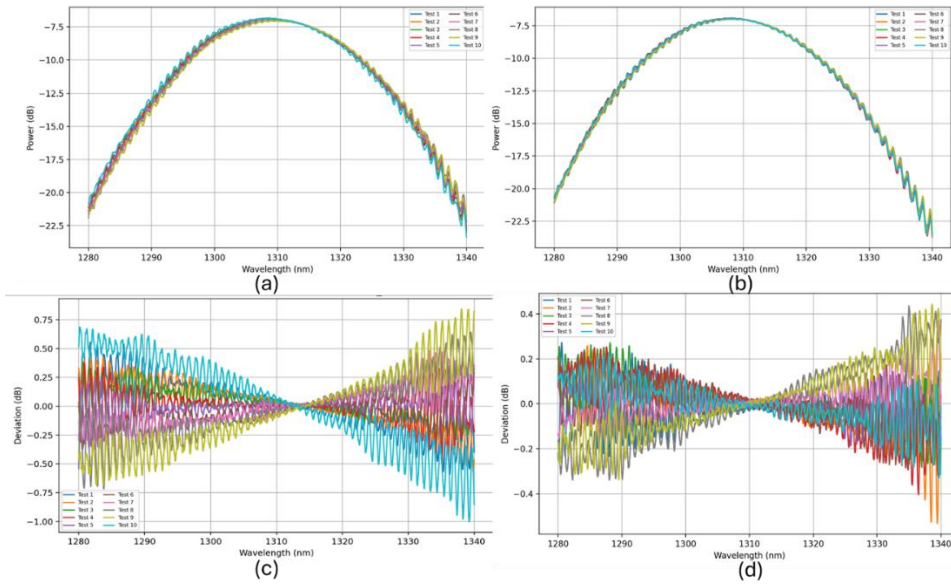


Fig. 1 Ten repeated wavelength sweeps of a single O-band grating coupler (1280–1340 nm). (a)–(b) Spectral overlays before and after optimization. (c)–(d) Corresponding power deviation from the mean spectrum before and after optimization. Post-optimization results show substantial suppression of sweep-induced variation.

Fig.1 shows the comparison of repeatability before and after mechanical and algorithmic optimization. Prior to optimization, peak power exhibited a mean of -6.904 dB with a standard deviation of $\pm 0.054 \text{ dB}$. For a $\pm 0.08 \text{ dB}$ tolerance window, this corresponds to a GRR of 87.5%. Peak wavelength deviation was $\pm 0.527 \text{ nm}$. After mechanical loop shortening, polarization stabilization, and curve-fitted peak extraction, peak power deviation decreased to $\pm 0.007 \text{ dB}$, yielding a GRR of 8.9%. Peak wavelength deviation was reduced to $\pm 0.087 \text{ nm}$. The optimized configuration satisfies OWAT repeatability requirements and suppresses sweep-induced drift.

3.2 Incident Angle Reconstruction

To quantify angular sensitivity, the effective beam incidence was reconstructed from measured 3D coupling trajectories.

For each die, the working distance Z was varied from $30 \mu\text{m}$ to $130 \mu\text{m}$ in $10 \mu\text{m}$ increments. At each Z position, an XY scan identified the peak coupling coordinate. The resulting 3D peak set $p_i = [x_i, y_i, z_i]^T$ defines a coupling trajectory. A total least-squares 3D line fit via principal component analysis (PCA) was applied to extract the direction vector $v = [v_x, v_y, v_z]$, from which angular components were computed using $\tan(\theta_x) = v_x/v_z$ and $\tan(\theta_y) = v_y/v_z$.

With $\pm 0.2 \mu\text{m}$ Z positioning accuracy over a $100 \mu\text{m}$ span, the corresponding angular uncertainty contribution is $< 0.03^\circ$, indicating that repeatability is dominated by peak-coordinate extraction rather than stage Z error.

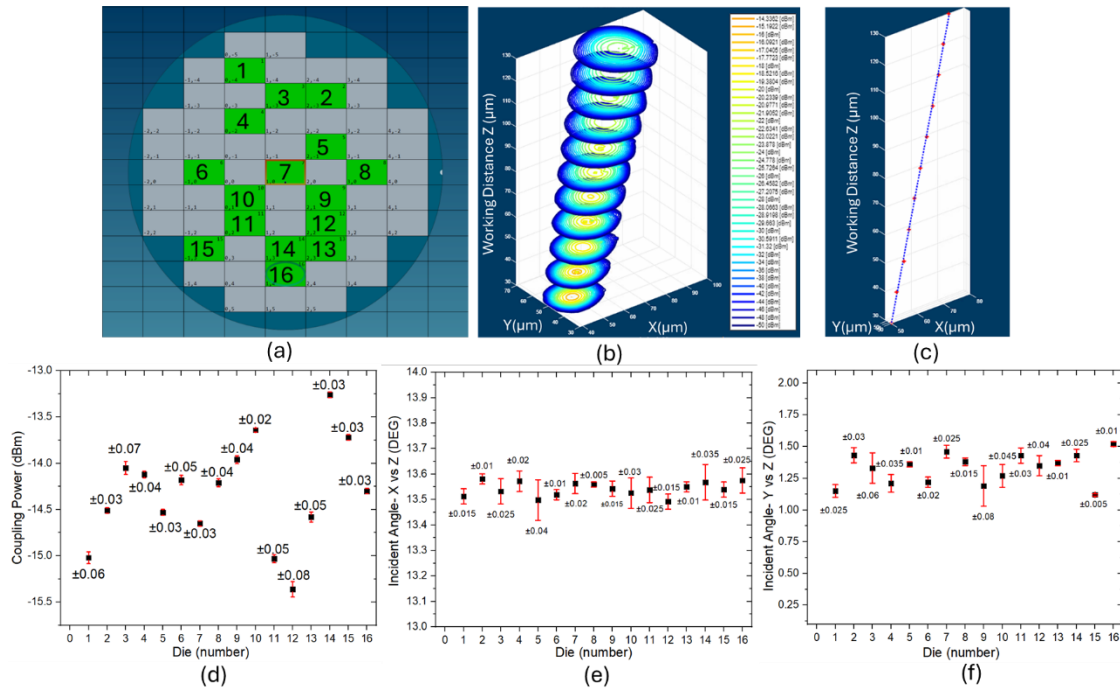


Fig. 2. (a) Spatial distribution of the 16 sampled dies across the 200 mm wafer. Fig. 2(b) presents a representative stacked 3D coupling surface obtained from multi-Z XY scans. The extracted 3D peak trajectory used for line fitting is shown in Fig. 2(c). Fig. 2(d) summarizes peak coupling power and repeatability for each die. The reconstructed angular components are shown in Fig. 2(e) for the X–Z direction and Fig. 2(f) for the Y–Z direction.

The dominant X–Z component is approximately 13.5° , with across-wafer variation within $\pm 0.1^\circ$ and repeatability per die between 0.01° and 0.08° . A non-zero Y–Z component ($\sim 1\text{--}1.5^\circ$) is observed with systematic die-to-die variation but strong repeatability, reflecting azimuthal FAU alignment and local wafer surface normal variation. Decomposing the incident direction into orthogonal components enables wafer-level angular mapping and improves alignment prediction during automated high-throughput testing.

4. Edge Coupler Characterization

Round-robin measurements were performed across 16 dies on a 300 mm wafer with V-groove structure. The full die sequence was repeated 20 times.

As shown in **Fig. 3(a)**, insertion-loss repeatability remains within $\pm 0.03\text{--}0.06$ dB, improving upon previously reported <0.2 dB performance [8]. Emission direction was reconstructed using the same 3D scanning method as shown in Fig. 3(b). The X–Z projection, Fig. 3(c) reveals a slight downward angular component consistent with fabrication tolerances. The results demonstrate stable automated EC testing with optional beam-direction characterization capability.

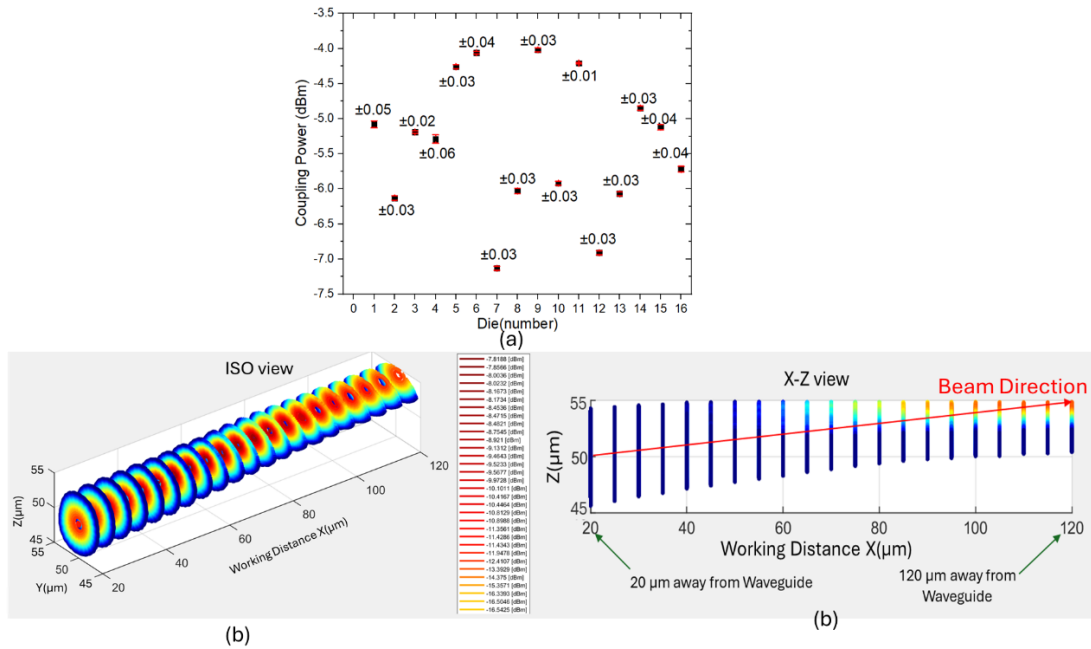


Fig 3. (a) Coupling power of 16 dies with 20 round-robin scans; error bars indicate repeatability per die. (b) Representative 3D coupling profile of the edge coupler output beam over a 20 μm working distance range. (c) X–Z projection of the reconstructed beam trajectory showing extracted emission angle.

5. Conclusion

A fully automated wafer-level optical probing platform for GC and EC characterization has been demonstrated. Mechanical and algorithmic optimization reduced GC wavelength-sweep GRR from 87.5% to 8.9%. A 3D trajectory method enabled wafer-level angular reconstruction. Automated EC testing achieved ± 0.03 – 0.06 dB repeatability, establishing foundry-grade wafer-level performance.

References

- [1] D. Thomson et al., “Roadmap on silicon photonics,” *J. Opt.* 18, 073003 (2016).
- [2] D. A. B. Miller, “Silicon photonics: Meshing optics with applications,” *Nat. Photonics* 11, 403–404 (2017).
- [3] M. Hochberg and T. Baehr-Jones, “Towards fabless silicon photonics,” *Nat. Photonics* 4, 492–494 (2010).
- [4] T. Taillaert et al., “A compact two-dimensional grating coupler used as a polarization splitter,” *IEEE Photon. Technol. Lett.* 15, 1249–1251 (2003).
- [5] L. Chen et al., “High efficiency apodized grating coupler for silicon-on-insulator waveguides,” *Opt. Express* 18, 18996–19003 (2010).
- [6] J. Notaros et al., “Ultra-efficient CMOS fiber-to-chip grating couplers,” in *Proc. OFC* (Optica Publishing Group, 2016), paper Th4A.3.
- [7] S. Lin et al., “Low-loss silicon edge coupler for broadband packaging,” *Opt. Lett.* 43, 3109–3112 (2018).
- [8] Q. Yuan et al., “Fully Automated Wafer-Level Edge Coupling Measurement System for Silicon Photonics Integrated Circuits,” *IEEE Trans. Semicond. Manuf.* 38, 168–177 (2025).